

PARTS LOGISTICS AUDIT PROGRAM #32-20-00

The release of this audit program to parties external to IBM requires the written approval of the Chief Auditor.

This program is intended as a guide for assessing risk management in this process through the performance of an audit or a similar engagement. Risk management is the process to identify, manage, and control potential events in order to provide reasonable assurance related to the achievement of IBM's business and control objectives. IBM's control objectives fall into five broad categories:

1. compliance with external laws and regulations, as well as IBM policies and procedures
2. accomplishment of business objectives
3. reliable information for financial reporting and decision making
4. efficient operations, and
5. safeguarding both intellectual and physical assets.

The control assessment questions are intended to address process objectives from the categories above and the risks to achieving those objectives. Objectives and risks in the actual process under review may differ based on organizational, operational, regulatory, and economic factors. As such, individuals performing the engagement must formulate a test strategy based on risks relative to the actual process under review, regardless of whether the risks are identified within this program.

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DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
Nov 2025	Self-Service Reports		Misook Cha	Delete this temporarily until audit can get complete access to extract the secondary control in Open Page
Nov 2025	Planning: Reference Material, Test Section		Misook Cha	Minor word update, Add full name of acronym
Sept 2025	Planning	Audit Program Risk Alignment	CA&AS Ops	Changes to audit risk levels from recent revalidation
June 2025	PLANNING	ALL	Misook Cha	Minor wording update
June 2025	PLANNING: Audit Program Risk Alignment TEST Sections		Misook Cha	Change the Primary Test to Secondary Test for IA003, SA001, SA002. Specify individual tests in RAE table accordingly.
June 2025	PLANNING: Data Analysis TEST Sections		Misook Cha	Update the Data Analysis from "Data mining" in all test sections Removed IDR, Data mining, Reference Material remark in each test for simplification & avoiding duplicated instructions.
June 2025	2. Parts Request Management 4. Inventory, Asset and Distribution Controls 5. System Access and Separation of Duties	Parts Request Management (PR001.P), Parts Supply Management (PP004.P) Inventory Controls and Verification (IA001.P) Separation of Duties (SA002.S)	Misook Cha	Add/Adjust testing approach
March 2025	Planning: Reference Material		Misook Cha	Updated the reference per the latest address
March 2025	4. Inventory, Asset and Distribution Controls	Inventory Controls and Verification (IA001.P)	Misook Cha	Independent Inventory counting testing approach was updated
November 2023	Planning: Data Analysis		Jan Aldrin Pe	Added notes for Self-Service Reports
February 2023	Planning: Data Analysis		Jan Aldrin Pe	Converted from Notes into a Word document and into a Box folder

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PROCESS AND OBJECTIVE

For the purposes of this audit program, field parts management is defined as the process of providing spare parts for the repair or service of customer or IBM owned equipment in a timely and effective manner. Included in the process are administrative procedures which ensure protection of assets and efficient customer service and financial procedures to account for the accurate statement of inventories. This audit program covers field parts from the planning and acquiring of sufficient parts, receipt by the ordering service representative, delivery of the parts to the accurate destination on time, and to the return of the parts. Inventory of the parts while in the custody of the service representative is also included. This process is often referred to as outside parts.

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PLANNING: Audit Program Risk Alignment

Parts Logistics

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Inventory and Controls	High	Inventory, Asset and Distribution Controls	IA001.P, IA002.P, IA003.S
Outsourcing Management	Medium	Outsourcing Management	OM001.P, OM002.P, OM003.P
Parts Planning	Medium	Parts Planning	PP001.P, PP002.P, PP003.P, PP004.P, PP005.P, PP006.P
Distribution Operation	Low	Inventory, Asset and Distribution Controls	IA004.S, IA005.S, IA006.S, IA007.S
Parts Request Management	Low	Parts Request Management	PR001.P
System Access and Separation of Duties	Low	System Access and Separation of Duties	SA001.S, SA002.S

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PLANNING: Data Analysis

Data analysis should be a part of every engagement where data is available in databases or data warehouses that can be accessed through query programs. This concept enhances the risk-based auditing approach, makes the audit process more efficient, and increases the productivity of the audit team. When the audit team does not have the skills required to perform these tasks, assistance should be sought from the Technology team.

All data used for analysis must be reconciled to an independent source so that the auditor has confidence that the universe is complete.

The [Data Analysis engagement procedures in the Audit Manual](#) (link internal to CA&AS members only) provide further suggestions.

This table shows what data can be reviewed and analysis techniques.

Data to be Analyzed	Analysis to be Performed	Concern
Metrics validation List of metrics provided by supplier	Were measurement reports executed as intended by the supplier? Were they properly validated by the central validation team? Did management have proper visibility to the metrics results? For those metrics that required action plans when targets were not attained, it will be validated if there was a root cause analysis (RCA), action plans defined and implemented.	Insufficient or incorrect metrics reporting could lead to: • Incorrect measurement on supplier's performance, • Errors remaining undetected and delays in actions to prevent financial loss and fraud. • Impair management abilities to take decisions impacting IBM's image and customer satisfaction.
Supply Order Management: List of unplanned / manual requisitions and emergency orders	Validate that: • Manual requisitions were approved • The orders were justified • check if parts numbers were in the list of reserved parts • There is a trend per parts numbers, PO, parts analyzer	Ineffective customer order management could result in missed customer shipments, customer dissatisfaction, and additional costs.

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Concern
Inventory Controls and Verification: Review list of inventory adjustments during the audit period	Validate that inventory adjustments were timely adjusted and properly documented. Determine if a pattern exists. (i.e. parts number, date of adjustment)	Provisions and manual adjustments not properly supported and performed untimely could lead to inaccurate financial reports, incorrect inventory level and financial loss.
Excess and Surplus: List of parts provisioned as Surplus, Excess and Obsolete	Analyze demand and purchases for these parts, prior to the month that was provisioned.	Inattention to excess, surplus, and obsolete parts could result in financial loss and customer satisfaction issues.
Sensitive Parts Management: List of sensitive parts variance	Review if unexplained negative variances exist to determine a pattern.	Possible theft.

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PLANNING: Reference Material

Planning Matrix/Scope Guide

The following matrix should be used as a guide when determining the scope areas to be reviewed when planning for an audit engagement, including: inventory management/distribution operations/cost accounting/procurement (if planned to be concurrent). During the preplanning review with the organization's management team, complete the following preplanning matrix guide.

Test Description	Global or Local	Site Level?	Outsourced or Other Org?
Inventory Planning	G + L		
Excess, Surplus & Scrap (ESZ) Management	Automated	Hubs	
Supply / Demand Management System			
Material Requirements Planning (MRP) and Supply Order Management	Global systems		
Supply Chain Specialist			
Replenishment			
Receiving			
Shipping			
Customer Order Management			
Customer Supply Availability			
Customer Satisfaction and Surveys			
Asset Protection			
Sensitive Parts Management			
Physical Security			
Inventory Controls			
Parts Movement			
Rotating Inventory Counts			
Rotating Inventory Audits			
Annual Physical Inventory			
Returned Parts Control			
Scrap			
Separation of Duties			

Asset Protection

- All below guidelines can be downloaded in the following site:
<https://w3.ibm.com/w3publisher/asset-protection>
 - CP 10. 12 Asset Protection – Excess & Surplus Parts Inventories
 - CP 10.13 Asset Protection – Normal and Sensitive Parts Security
 - CP 10.15 Asset Protection – Returned Parts MES and EC Process Requirements

Inventory Verification

- AI-INVCA00-005 Rotating Inventory Audits of Inventories Conducted by Accounting:
<https://ibm.ent.box.com/s/lixo7vvsi3u2b319rzg33985zli3eu79PLANNING: Reference Material>
- AP INV-VER- Inventory Verification: <https://ibm.ent.box.com/s/zzvtthvvv8h3vamukwccslndbg7q91r>
- AP INV-CON- Consignment Inventory Accounting and Controls:
<https://ibm.ent.box.com/s/q566sy0guxwfnr94sk8bc0otpj2malre>

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PLANNING: Reference Material
(Continued)

Labelling

- CB03700000 Tamper Evident Packaging:
<https://ibm.ent.box.com/s/s7bzbmjsqfkmcxousnm9ijuds3rvkmu9>

Scrap

- Scrap Process:
<https://w3.ibm.com/w3publisher/global-fixed-asset-management-wiki/my-asset/asset-disposal/scrap>

System Access and SOD

- IT Security Standard (ITSS): Please refer to the section 4.0 Access Controls and 6.0 Operations Security in IT Security Standard (ITSS) managed by IBM Cybersecurity – PSG.
<https://pages.github.ibm.com/ciso-psg/main/standards/itss.html>
- Fin 182 Management of Self-Assessment/Appendix C SOD:
<https://w3.ibm.com/w3publisher/financebc/ibm-framework/ci-fin-182-appendix-c>

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PLANNING: Audit Protocol - PLEASE READ!

Protocol for Auditing in an Outsourced Environment

As part of the divestiture of IBM Global Logistics (GL), the Lead Logistics Provider (LLP) supplier has assumed direct responsibility for IBM's physical logistics needs, becoming IBM's LLP outsource supplier of logistics services.

In support of the outsource contractual relationship post divestiture; the following audit protocol is to be implemented within the 3 core GL Audit Programs below:

- Parts Logistics, 32-20-00
- GL - Physical Logistics, 32-10-00
- GL - Trade Compliance, 32-30-00

This protocol is to cover formal reviews, inspections, and or audits with the intent to provide both companies a framework to conduct these engagements in an efficient and productive manner.

These engagements are conducted with the intent to determine the controls health of key processes and to implement process improvements to assure and sustain a strong controls posture. This is beneficial to both companies to meet an environment of increasing regulatory and legal requirements.

In some cases, Lead Logistics Providers (LLPs) use and manage a network of suppliers that are contracted to IBM. In other cases, described as LLP below, the LLP uses and manages their own network of suppliers. LLP suppliers have assumed direct responsibility for IBM's physical logistics needs, becoming IBM's Fourth-Party Logistics (4PL) Suppliers. LLP suppliers have also assumed the management of IBM's existing global logistics contracts which are now Third-Party Logistics (3PL) suppliers of logistics activities to IBM. While LLP suppliers are IBM's partners, they are separate legal entities, not owned by IBM.

All contracts with service providers where IBM Global Logistics Procurement was previously the buyer have been transferred to LLP suppliers (few exceptions only). IBM Global Logistics no longer has a direct contract responsibility with LLP sub-tier suppliers. The contract responsibility with the 3PL service providers is now owned and managed by LLP suppliers.

Primary point of contact for reviews, inspections, and or audits is with the IBM Control Tower representative.

The standard notification from IBM to LLP suppliers will be 30 days in advance of review. There will be circumstances; however, where the schedule will be dictated by other factors, such as preparation for a short notice customs audit, resulting in less than 30-day notification.

Establish an LLP supplier point of contact for specific engagement. LLP supplier will assign a company representative to speak on behalf of the company in discussions. When requests fall into LLP area of responsibility the IBM Control Tower representative will involve the designated LLP interface.

As a practice, requests of LLP suppliers or sub-tier suppliers should be focused on results and not with the supplier owned processes, procedures or work instructions. Because of IBM's regulatory and legal risks/liability, requests for data and observation of LLP supplier's or sub-tier supplier's security measures and certain processes at an execution or activity level will be generated for trade compliance reviews.

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PLANNING: Audit Protocol - PLEASE READ!
(Continued)

LLP suppliers will establish a management system to support requests originated by IBM. The LLP supplier interface will involve the process owner in LLP supplier sub-tier suppliers as required. LLP suppliers will use reasonable effort to support requests within 24 hours.

Reports (formal audits or other IBM internal reviews) will not specifically identify LLP suppliers by name but will refer to them as the "supplier/vendor or any other appropriate designation". However, IBM reserves the right to identify LLP suppliers by their corporate names when necessary and will consult Legal about any circumstances that this may be a requirement, e.g., a trade compliance audit. The IBM Control Tower representative will validate and confirm all facts related to LLP supplier executed processes with them prior to signing or agreeing with the IBM internal reviewer's/inspector's/auditor's results.

Resulting findings, recommendations, and actions directed at the sub-tier supplier should be addressed using the current management system and, if needed, the Change Management Request (CMR) process established in the Statement of Work (SOW). This should be focused on the results with the sub-tier supplier implementing those actions they deem necessary to deliver the expected results.

Definitions:

4PL – A fourth-party logistics provider (abbreviated 4PL), is a consulting firm specialized in logistics, transportation, and supply chain management. As such, a 4PL is an integrator that assembles the resources, capabilities, and technology within its own organization and other organizations to design supply chain solutions. 4PLs use 3PLs to supply service to customers, owning only computer systems and intellectual capital to do so.

3PL – A third-party logistics provider (abbreviated 3PL, or sometimes TPL) is a firm that provides a one stop shop service to its customers of outsourced (or "third party") logistics services for part, or all of their supply chain management functions. 3PL providers typically specialize in integrated operation, warehousing and/or transportation services that can be scaled and customized to customer's needs based on market conditions and the demands and delivery service requirements for their products and materials.

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PLANNING: Fraud Template

Fraud is any intentional act or omission designed to deceive others and resulting in the victim suffering a loss and/or the perpetrator achieving a gain. Fraud can be categorized into fraudulent financial reporting, misappropriation of assets, and improper or unauthorized expenditures.

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Theft of sensitive parts	Physical security, physical inventory, separation of duties. Unexplained negative variances over tolerance should be reviewed with the local security organization.	Review physical security and test warehouse access control. Review results of physical inventory counting, including approval of write-off. Review separation of duties assessment against actual practice.

Note: The probability and impact will vary depending on the organization/business unit that is being audited.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Parts Planning

Objective:

Is the parts planning process effective to ensure that supplies meet customer demand through appropriate planning, while balancing the objective of supply availability with inventory costs? Is there synchronization of demand uncertainty with supply flexibility through management and accurate planning of lead times? Are there reliable measurements in place to gauge the effectiveness of the process?

Risk/Exposure:

An ineffective planning and measurement process can result in poor customer satisfaction due to missed customer shipments and non-attainment of business objectives due to excess cost.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Parts Planning

Primary Test: Policy Inventory Planning (PP001.P)

Objective:

Is the policy inventory planning process effective to ensure balancing the objective of supply availability with inventory costs? Are there reliable measurements in place to gauge the effectiveness of the process?

Approach:

Achieving Business Objectives

An ineffective policy inventory planning process can result in missed business commitments and excess cost.

Testing Strategy

Obtain copies of all inventory measurements for the twelve months of the audit period. Review the measurements for appropriate targets, performance trends, and action plans.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Parts Planning

Primary Test: Supply Demand Planning (PP002.P)

Objective:

To determine if the supply/demand management system is effective and is consistent with demand/supply planning cycle. Is there synchronization of demand uncertainty with supply flexibility through management and accurate planning of lead times?

Approach:

Process Risk

Achievement of Business Objectives: Incorrect demand could result in erroneous supply commitments and excessive cost.

Testing Strategy

- Is there evidence that all sources of inventory are included in the plan:
 - Parts ordered from IBM manufacturing
 - Parts ordered from suppliers
 - Parts in stock, including New, Certified Service Part (CSP), and Available for Repair (AFR)
 - Parts outstanding for return
- Have inventory exposures been highlighted in the supply decision process?
- Determine if the process has flexibility to adjust based on demand fluctuation.
- Determine if demand adjustments have been approved and communicated appropriately.
- Is there assessment of supply availability of critical parts?

Customer Satisfaction

Inaccurate demand could result in customer commitments not being met.

Testing Strategy

- Review demand input and determine if demand is unbiased? How is the basis of the data determined? Is the basis of the data validated?
- Is there evidence of management approval?
- Have key supply/demand process indicators been deployed for the planning process? Are measurements and actions in place for non-attainment and reviewed with executive management on a regular basis?
- Were the shipping commitments updated as a result of the analysis and management reviews?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Parts Planning

Primary Test: Parts Requirements Planning (PP003.P)

Objective:

To determine that controls over material requirements are generated to produce accurate order recommendations.

Approach:

Process Risk

Customer Satisfaction and Achievement of Business Objectives

Inaccurate parts planning can result in parts shortages, inventory overages, and missed shipments.

Testing Strategy

- Review parts planning validation to ensure that committed demands are properly loaded into the planning application.
- Request list of miscellaneous requirement inputs to the planning application and select samples and test that they are properly supported and reviewed.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Parts Planning

Primary Test: Parts Supply Order Management (PP004.P)

Objective:

To determine, based on the planning process, if orders for parts are properly placed and managed. In addition, were there adequate controls and reviews for identifying, processing, and resolving excess, surplus and scrap inventory along with timely implementation of engineering changes.

Approach:

Process Risk

Customer Satisfaction and Achievement of Business Objectives

Inaccurate supply management can result in parts shortages, inventory overage, or missed shipments.

Testing Strategy

- Review orders based on products that did not meet scheduled commitments, and/or have significant inventory overages or shortages. Test for the following:
 - Execution of requirements generation recommendations.
 - Actions taken to monitor and control short-lead-time orders and firm orders in excess of lead-time.
- Verify reconciliation of open orders between locations that are tracked and aged. (Aged items are resolved timely.)
- Verify that open purchase requests are aged and tracked. (Aged items are resolved timely.)
- Review cancellation charges and check to see if root cause analysis was performed and that proper approvals were in place.
- Ensure materials that were identified as critical, short parts, and on back-order, are aged, reviewed, and resolved timely.
- Request list of unplanned/manual requisitions. Select a sample and test for:
 - Reason for manual requisition
 - Measured, tracked, and properly reviewed. Validate that the unplanned/manual requisitions are measured (should be a report) and tracked (reviewed and validated why they are existing and whether they are necessary/applicable).
- Request supply order list & receipt list. Select sample and test for:
 - Aging: Focus on the ones that are past their due date and still not received.
 - Receipt discrepancies: Ask for received orders deviating from the requested quantity and delivery date
 - Aging Resolved Timely: Activity is being reviewed timely if the activity is in control and an escalation process is in place in case it takes too long, etc.
 - Parts from external suppliers: For the parts manufactured by external suppliers, there is a contractual delivery time and a described escalation process when higher level management should be involved when timely resolution is not reached. The escalation process should be in the guideline.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Parts Planning

Primary Test: Parts Supply Order Management (PP004.P)

(Continued)

- Review Billed-Not-Received (BNR)/RNB and Invoiced-Not-Received (INR)/RNI activity for audit period. Determine if excessive, verify management involvement, verify current action plans are in place to resolve issues.

Note: Billed not received (BNR) refers to goods that have been billed from an internal IBM vendor and not received by the plant. Invoiced not received (INR) refers to goods that have been invoiced from an external vendor and not received by the plant. The RNB/RNI relationship is the same as above with goods that have been received by the plant and not billed or invoiced. The excessive aging of all four of these (BNR/RNB and INR/RNI) are control concerns, because IBM cannot be confident that cost and inventory is properly stated if these have not been reconciled.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Parts Planning

Primary Test: Returns and Reutilization (PP005.P)

Objective:

To determine if monitoring of re-utilization suppliers is in place to avoid misuse of IBM assets and ensure adequate cost analysis is performed to allow right buy or repair decision.

Approach:

Process Objectives

Generate the parts reutilization when replacement is needed for the customer and optimize usage of the parts returned.

Process Risk

Insufficient monitoring over re-utilization suppliers could impact the inventory planning, and lead to excess of inventory and parts lost. Also, could result in customer dissatisfaction and impact IBM's image. Parts not returned to IBM warehouse when required could expose IBM assets to misuse.

Inadequate analysis of cost to determine whether to buy or repair could result in financial loss.

Testing Strategy

- Ask for all requisitions for repair emitted between IBM and the repair suppliers during the audit period.
- Ask for the list of all parts bought and repaired during the period.
- Ask for the list of manual orders.
- Validate that an adequate comparison of cost was made to determine to repair parts or purchase new parts.
- Verify that the used parts returns are tracked and returned in a timely manner.
- Verify that repair orders are executed on a timely manner by the supplier.
- Ask for the measurement available associated to this process.
- Verify that penalties were applied, if applicable, when parts repaired were returned to IBM later than the agreed delay.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Parts Planning

Primary Test: Excess Surplus and Scrap (ES&Z) (PP006.P)

Objective:

Are there adequate controls and reviews for identifying, processing, and resolving excess, surplus, and scrap inventory?

Approach:

Process Risk

Noncompliance to Corporate Instructions CP10.12 and CP10.13.

Testing Strategy

- Determine if an operating procedure and organization exists to support excess surplus and scrap (ES&Z) management principles as defined in CP10.12.
- Determine that metrics and corporate measurements are accurate, reported timely, and problem areas are highlighted with corrective action plans in place. Reference CP10.12 (monthly reports).
- Determine that the ES&Z code definitions are adhered to. If not, has the division controller approved of the business rationale and notified the CHQ director of Integrated Supply Chain (ISC) processes and operations? Reference CP10.12.
- Verify that blocking codes for site declarations are utilized. Reference CP10.12.
- Determine that ES&Z is recognized as preferred supply and that vendor terms and conditions provide maximum flexibility to allow optimum ES&Z usage. Reference CP10.12.
- Determine that parts recommended for scrap are first reviewed for potential external sales opportunities in accordance with CP10.13.

Achievement of Business Objectives

- Determine that the site is utilizing all available tools and reports to minimize ES&Z inventory. Examples of tools and reports are: the PSG Impact tool, WWW inventory tool, and the interplant logistic system (IPLS) weekly declaration, missed opportunity, and high value reports.
- Determine if multi-level-balancing opportunities exist and if a management system is in place to maximize the multi-level balancing opportunity prior to sale/scrap/salvage operations. Note: Balancing means that if there is overage inventory at one IBM location and a shortage at another IBM location that the inventory is offered by the overage location and ordered by the shortage location to avoid scrap at the overage location and buying externally at the shortage location. Multi-level refers to the fact that balancing can happen between SPO locations within a geography (level 1), between SPO geographies (Americas, EMEA, AP (level 2)) and across business units (SPO, GARS, manufacturing).
- Loss of assets and fraud being undetected.
 - Determine that the separation of duties (SOD) supports the basic ES&Z management principles. Request a copy of the ES&Z SOD assessment and review for conflicting job responsibilities for ES&Z personnel. Reference CP-10.12 (management principles and responsibilities).

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Parts Request Management

Objective:

Did management have an effective process for planning and fulfilling Parts Requests in accordance with customer requirements?

Risk/Exposure:

A defective process for handling requests for parts could result in missed customer shipments and excess cost.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Parts Request Management

Primary Test: Parts Request Management (PR001.P)

Objective:

To verify that parts requests were being fulfilled appropriately.

Approach:

Process Risk

- Parts Availability: The consequence of any or all of these factors could lead to IBM having dissatisfied customers if the parts were not available as required to execute maintenance services.

Testing Strategy

- Have metrics been established to track and monitor the timeliness and accuracy of parts requests? If so, review metrics attainment and action plans for metrics that were not achieved.
- Verify that emergency parts orders were managed timely and properly with relevant approvals. Management also monitor that the list to ensure that the reason was justified and parts planning is still reasonable.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Outsourcing Management

Objective:

To determine that line management has implemented a process to ensure that the requirements of the vendor's statement of work (SOW) are being met and established a method for the validation of services received. In addition, to determine if payments to vendors are in accordance with contractual terms and conditions, measurements are in place to monitor the vendor performance, assets are being safeguarded, and procurement is involved in the maintenance of the contract and supplier performance reviews.

Risk/Exposure:

If management had not established a process for the verification of receipt of goods and services, then payments could be made for services not received. Contract terms and conditions, including a definition of the scope of work, must be clearly stated in the contract, or prolonged disputes with vendors can result which could impact the continuity of services. Failure to adequately control IBM assets at vendors could result in supplier relations problems, loss of IBM assets, diminished customer relations, or noncompliance with government regulations. In addition, if proprietary information was not carefully controlled, IBM could lose advantage in the marketplace. Further, failure to ensure that the amount declared to customs is the price paid or payable plus any specific additions resulting from supplier contracts could result in inaccurate declarations and payment of incorrect duties and taxes leading to fines, penalties and/or back-owed taxes with interest payments.

Failure of suppliers to meet established performance targets decreases IBM's ability to provide high-quality, low-cost products and services to our customers. Additionally, the absence of a good internal measurement system creates an environment for fraud to occur and can lead to excessive expenditures.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Outsourcing Management

Primary Test: Scope of Work Management (OM001.P)

Objective:

To determine if line management has established processes and procedures to manage the scope of work assigned to vendors. Determine if contracts were established utilizing procurement, included detailed scopes of work, appropriate business and legal clauses, and defined supplier performance measurements. Also, to determine if contracts were adequate to deliver the business objectives of the outsourced relationship.

Note: If the outsourcing management organization is not a procurement organization, was the contract negotiated and issued by a procurement organization?

If the organization being audited has delegated procurement and contract management responsibilities, were these roles clearly defined and understood?

Approach:

Process Risk

If management had not established a process for the verification of receipt of goods and services then payments could be made for services not received. Contract terms and conditions, including a clear definition of the scope of work, must be clearly stated in the contract, or prolonged disputes with vendors can result which could impact the continuity of services.

Testing Strategy

- Review the process documentation developed by the line organization that defined the management responsibilities for meeting organizational business objectives through outsourced manufacturing. Verify the following:
 - Process measurements had been established to identify deficiencies by the vendor in performing the vended services.
 - A process for updating contracts and agreements had been established and agreed to by the vendor.
 - A process had been established to verify that prices do not exceed contracted amounts. If prices were not stated in the contracts, was there a process for verifying that the prices were consistent with industry standards.
- Review the outsourced contract(s) for the following criteria:
 - Did the scope of work clearly define the services to be performed by the vendor? Were these requirements complete, deliverables defined, and measurable?
 - Were supplier performance measurements specified in the scope of work? Had line management established a method for validating that services had been received or performed?
 - Was there a process to track and monitor contract deliverables to assure realization of business objectives?
 - If the audit clause had been exercised, were the results reviewed and appropriate action plans implemented?
 - Were change control procedures established so that line management could request changes and updates to the scope of work? Review recent changes to the scope of work to determine that the established process was followed.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Outsourcing Management

Primary Test: Scope of Work Management (OM001.P)

(Continued)

- Did the contract define the importer and exporter of record for international shipments?
- Did the contract contain clauses regarding import/export compliance requirements such as country of origin marking and product classification?
- When PO with suppliers include additions to the unit price, such as periodical payment, communication of these payments to importing country global logistics for declaration consideration. Does the process effectively ensure that additional payments to suppliers are identified and communicated to Global Logistics?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Outsourcing Management

Primary Test: Assets at Suppliers (OM002.P)

Objective:

To determine if assets loaned to non-IBM entities were adequately controlled to prevent supplier relations problems and the loss of IBM assets.

Approach:

From the location's property control system and from the location's loan agreement tracking system, obtain a listing of all assets identified as being located at a supplier's site. Select a sample for review. In addition, determine if consigned parts were provided to vendors, select a sample for review.

Testing Strategy

For the sample selected, test against the following criteria:

- Was an equipment loan or tool loan agreement in place?
- Did the supplier sign the agreement and acknowledge receipt of the equipment?
- Was a physical or paper inventory performed, as required?
- Was the asset properly returned, if applicable?
- When assets and equipment are provided to the supplier it must be communicated to Global Logistics for declaration:
 - Does the process effectively identify goods on consignment at supplier locations for reviewed by Global Logistics?
 - When PO with suppliers include additions to the unit price, such as periodical payment, communication of these payments to importing country global logistics for declaration consideration.
 - Were controls effective to ensure the timely communication to global logistics? This review can be part of a global or regional process as agreed with Global Logistics. Does the process effectively ensure that additional payments are identified and sent for review by Global Logistics?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Outsourcing Management

Primary Test: Supplier Performance Management (OM003.P)

Objective:

To determine if measurements were in place and functioning which provided management visibility to supplier performance against the requirements specified in the contract and scope of work.

Approach:

Testing Strategy

Review line management measurements to determine if supplier performance criteria for procured goods and services was monitored and reported to management. Evaluate the management process for determining that performance measurements were achieved.

- Had measurements been established, including reporting frequency, management review of reports, and reporting to determine that required deliverables specified in the scope of work were being received?
- Do measurements focus on timeliness, quality, quantity, and cost?
- Do these measurements include all requirements in the scope of work and other contract terms?
- Is the vendor providing timely and accurate measurements to IBM as required in the contract?
- Were supplier audits conducted and if necessary, were corrective actions taken based on the results?
- When supplier performance targets were not met, were there documented action plans developed?
- Line measurements of supplier performance are being shared with procurement management as input to the procurement supplier performance reporting for core suppliers?
- If the contract defines performance guarantees, were they invoked when performance levels were not met? Verify that unsatisfactory situations were being improved through the use of action plans.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory, Asset and Distribution Controls

Objective:

Does management have an effective process to safeguard assets (in inventory) both physically and logistically?

Risk/Exposure:

The lack of proper controls over the security of IBM assets can lead to lost or stolen inventory.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory, Asset and Distribution Controls

Primary Test: Inventory Controls and Verification (IA001.P)

Objective:

Did management have an effective process to verify that controls were in place to ensure that the IBM inventory was correctly accounted for?

Approach:

Process Risk

- Asset Loss: If processes were not effective to verify inventory, asset losses may go undetected and financial records could be overstated.
- Undetected Fraud: An individual with job responsibilities and/or system access capability which would permit the initiation and concealment of irregular activity could make fraudulent payments which result in financial loss to IBM.
- Missed business objectives: In addition, inaccurate inventory records could negatively impact the requirements generation process, which relies on these records to calculate order points and quantities.

Testing Strategy

- Determine that (rotating) inventory counts (both RIC and 100% counts) were performed and that all inventories were included. Test against the following criteria:
 - Identify who set the parameters that determined which parts would be counted.
 - Review the inventory count results to determine if variance levels were calculated correctly and that the variances were in compliance with parameters.
 - Review if inventory variances were properly analyzed and inventory adjustment requests were approved prior to processing.
- Verify that duties were separated among generating the list to count, counting, recounting, system access, investigations, and adjustments.
- Conduct independent inventory counting (book-to-wall, wall-to-book) and verify if warehouse environment is in line with the vendor contract as well as IBM's policy. Please refer to the counting sheet sample "*Template WH Inspection_v0.xlsx*".

PARTS LOGISTICS

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory, Asset and Distribution Controls

Primary Test: Sensitive Parts Management (IA002.P)

Objective:

To determine if the Control Tower's control processes, data accuracy validation, and documentation of LLP related to sensitive parts are adequate in protecting them from loss or theft. In addition, that the loss of sensitive parts is recognized and reported per the Master Statement of Work Logistics Services Supply Agreement.

Approach:

Process Risk

Noncompliance to the Master Statement of Work Logistics Services Supply Agreement could result in lost or stolen inventory.

Testing Strategy

Validate that the Control Tower has done the following:

- Ensure that the Control Tower has sufficiently validated and controlled the LLP supplier's review and validation of the input from 3PL suppliers which demonstrates that all sensitive parts received at the site during the audit period can be tracked from place to place as required by their APC classification.
- Ensure the Control Tower has validated LLP supplier's verification of 3PL suppliers on open APC4 shipment report for timely closure and escalation where required.
- Ensure LLP supplier has verified that the 3PL supplier has reviewed open APC4 receipt report for timely closure and escalation where required.
- Ensure LLP supplier has reviewed and validated a listing of all lost sensitive parts during the audit period.
 - LLP supplier has verified that all lost Asset Protection Code 4 (APC) parts were timely identified and reported promptly to management.
 - LLP supplier has verified that all lost APC 4 parts were accurately entered into the buyers asset tracking system (CATS) per the buyer's requirements.
 - LLP supplier has verified detection of loss and correction of any out of tolerance conditions that may exist and appropriately reported all instances to Security, line and financial management where variances exist above the approved levels in accordance with Corporate Financial guidelines.
- Ensure LLP supplier has verified from a report, all write-offs. Test for any unexplained negative inventory variances out of tolerance. After an investigation has been completed, ensure that the results are reviewed with the local security organization for the possibility of theft or fraud as documented by the unit's management approved reconciliation process.
- Ensure LLP supplier has verified that frequency of cycle and inventory counts for sensitive parts are based on the buyer's requirements.
- Ensure LLP supplier has verified that test compliance to tamper evident sealing are based on buyers requirements.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory, Asset and Distribution Controls

Secondary Test: Environmental Management (IA003.S)

Objective:

To determine if processes are in place and controls are defined to support IBM's corporate environmental policy, instructions, practices, and standards and if procedures are in place to ensure compliance.

Approach:

Process Risk

Noncompliance to IBM's corporate environmental policy may result in penalties or fines and into publicity adverse to IBM.

Testing Strategy

- Validate that:
 - The Technology Lifecycle Service (TLS) Enterprise Architecture (EA) Supply Chain (SC) Regional Coordinator is in place and participates in all meetings applicable to TLS SC.
 - The TLS SC Chemical Coordinator is the same person as the EA coordinator and that a process is in place to handle (hazardous) chemicals.
 - Waste parts shipments are prepared in line with the regulations.
 - IBM instructions and applicable legislation are executed upon.

Noncompliance to IBM's corporate environmental instructions, practices and standards may result into injury, harm or destruction of personnel, facilities and the environment.

- Validate that:
 - Processes and procedures for the control of chemicals and chemical safety have been effectively implemented and are in compliance with IBM regulations and legal requirements.
 - Processes and procedures are in place for reviewing, approving, storing, distributing, or selling of chemicals and that they are in accordance with IBM and legal requirements.
 - A chemical coordinator has been assigned to manage and ensure compliance of these processes.

Noncompliance to IBM's corporate environmental standards with regards to the Multi-Vendor Services (MVS) parts business may result into penalties or fines and into publicity adverse to IBM.

- Validate that:
 - There is a process in place for ensuring compliance with environmental regulations and IBM requirements affecting the distribution and /or sale of service parts, including those manufactured by others, including, but not restricted to substance restrictions, energy conservation and efficiency, battery management, transportation of dangerous goods and other environmental topics.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory, Asset and Distribution Controls

Secondary Test: Supply Chain Security (IA004.S)

Objective:

To validate if IBM's Supply Chain Security (SCS) commitments have been incorporated into all global Supply Chain (SC) distribution facilities. Specifically, to determine that an annual assessment of IBM's compliance with supply chain security requirements and Authorized Economic Operator (AEO) program requirements has been performed and any corrective actions required have been implemented.

Approach:

Review the process to determine the following:

- What was the result of the most recent supply chain security assessment (Import Compliance Office (ICO) Supply Chain Security Checklist) and/or Supplier annual self-assessment security checklist?
- What controls/tests are used to identify supply chain security deficiencies?
- How are identified deficiencies in the supply chain security assessment tracked for action plan development and closure?
- If the location participates in a government sponsored supply chain security program (e.g., AEO), were annual profile reviews and updates performed and communicated to the government within required timeframes? (In the U.S., the program is known as Customs-Trade Partnership Against Terrorism (C-TPAT). Although controls to prevent the introduction of items (terrorists, weapons of mass destruction, narcotics, money) into our supply chain are embedded throughout the enterprise, at a manufacturing or distribution center location you will be able to observe controls implemented as shipments are created and trucks/trailers/containers are stuffed with IBM products for international transportation.)

Location

Assessments and Supply Chain Security checklists can be reviewed in the central location for all locations/countries. The observation of the execution (primarily stuffing of containers destined for transport) needs to be performed at shipping locations.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory, Asset and Distribution Controls

Secondary Test: Customs Free Zones (CFZ) (IA005.S)

Note: This test covers all operations in a Customs Free Zone (CFZ), which includes all areas/zones under Customs supervision that are generally considered outside Customs territory for the purposes of import duties and tax only, or within national borders but where the trade laws are different and special privileges are granted. These areas include Foreign or Free Trade Zones (FTZ), Special Economic Zones, Software Technology Parks, Export Processing Zones, Bonded Warehouses, etc. A CFZ is a secure area operating under the supervision of customs authorities. CFZs are generally used to defer payment of duties until merchandise enters the commerce of the country or the merchandise is shipped to another country. Manufacturing CFZs are generally single-purpose sites operating as a subzone of the grantee because the general-purpose zone cannot accommodate the manufacturing process. Merchandise in the manufacturing zone can be manipulated, manufactured, destroyed, exhibited, or temporarily removed with the proper permits. Even when CFZ operations have been discontinued, it is important to validate that appropriate records are retrievable for the full period required under the statute of limitations or specific regulations, whichever is longer.

Objective:

To validate that internal controls ensured the compliant storage and processing of Supply Chain (SC) merchandise admitted into a CFZ.

Approach:

Validate the following:

- Are there any SC locations which operate in a FTZ or CFZ including Bonded Warehouse?
- What controls are in place to ensure that IBM inventory is managed in accordance with Customs requirements for FTZ/CFZ?

There is no Free Trade Zone Operation as of June 2025. A discussion is on-going to have a FTZ in US. Below is for the FTZ if it becomes active.

Not all countries will have operations in CFZs, but most will have operations in bonded warehouses. It is a test that can also be performed remotely (although it would be helpful to visit), but the team can, e.g., validate the inventory reconciliations vs. the declarations to customs for discrepancies. The terms and conditions will vary by country, but the audit team could determine whether we are compliant with the contracts. It is a key test, and the audit team should request an overview of customs free zones for the process reviews.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory, Asset and Distribution Controls

Secondary Test: Invoice Quality (IA006.S)

Objective:

To determine that declarations to customs and other government agencies were accurate, adequate, and timely and corrections, if permitted, were made when required. As the commercial invoice is used for the customs declaration, the purpose of this test is to ensure that specific requirements related to content, language, and layout are met and that quantity, weight, and dimensions are accurate.

Approach:

Validate the accuracy of source information on the invoices related to product cost value, description of items, and origin within processes which produce them.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory, Asset and Distribution Controls

Secondary Test: Import Licenses, Permits, Certificates, Notifications, and Approvals (IA007.S)

Note: A small number of countries introduce trade barriers to restrict the import of specific products (such as batteries and products with encryption) through the introduction of special requirements for licenses, permits, certificates, notification or preapproval by authorities either prior to or at the time of import. Note: this also applies to used goods, which is tested separately. These controls may be implemented for reasons of national security (as is with the case for encryption controls), and/or to protect the local economy or environment. Failure to comply with these controls may result in civil or criminal penalties, or the loss of IBM's privilege to import and deliver products and services to these countries.

Objective:

To determine that any local requirements were met.

Approach:

Validate:

- How are countries with these specific requirements determined in Supply Chain (SC) warehouse before shipping?
- Review country prohibitions and, for restricted products, requirements for any import licenses, permits, certificates (i.e., battery, etc.), notifications or approvals that are needed. This includes Used Goods.
- How are Used Goods physically and on the system identified from new stock in the inventory?
- Were the conditions of the specific restriction (license, permit, certificate, etc.) adhered to?
- Were local requirements for reporting and record-keeping met?

PARTS LOGISTICS AUDIT PROGRAM #32-20-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. System Access and Separation of Duties

Objective:

Are there adequate access, update capability controls, and separation of duties to prevent unauthorized changes to programs or fraudulent transaction processing?

Risk/Exposure:

Separation of duties involves the assignment of responsibility for a transaction in a manner such that the duties of one employee automatically provides a cross-check of the work of one or more other employees. The primary purpose of separation of duties is the prevention and prompt detection of errors or irregularities in the performance of assigned responsibilities. Functions are considered to be incompatible from a control standpoint when it is possible for an individual to commit errors or irregularities in the normal course of their duties and to prevent their detection by the system. Individuals with incompatible job responsibilities and system access capabilities are an exposure to IBM.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. System Access and Separation of Duties

Secondary Test: System Access Controls (SA001.S)

Objective:

Verify that management controls and authorization of all access to the application and data can be demonstrated with management awareness and removal of such access is processed when there is no longer a business justification (e.g., at separation or change of assignment); a valid business need for privileged user-ids exists and is current; accountability for shared user-ids can be established; and that auditable records are maintained to show compliance with IBM security standards.

Approach:

Process Objectives

IBM's security strategy is to grant or deny access to services and information, based on authenticated identity, individual business need, and on broad entitlements.

Process Risk

Data Integrity - Unauthorized persons may gain access to the application and its resources enabling them to enter unauthorized transactions or to make unauthorized changes to data or programs.

Testing Strategy

- Select sample of user-ids and test for authorization and revalidation.
- From a universe of users who have left the company, select a sample of requests to revoke access that were sent from a manager to the application security administrator. Verify that the user-ids were removed or disabled in a timely manner.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. System Access and Separation of Duties

Secondary Test: Separation of Duties (SA002.S)

Objective:

To verify that no one individual had incompatible responsibilities that would allow the system of internal controls to be circumvented and to ensure employee's responsibilities, system access, and physical access to inventory are properly segregated.

Approach:

Process Risk

Undetected Fraud - An individual with job responsibilities and/or system access capability which would permit the initiation and concealment of irregular activity or fraudulent payments, resulting in financial IBM loss.

Testing Strategy

- Review the IBM SOD assessments. They should include job responsibilities, physical access to inventory, and all critical application accesses, and be regularly reviewed and validated.
- Ensure that for SOD assessments conflicts were appropriately identified and resolved.
- If approved risk acceptances were in place to address SOD issues, ensure that the secondary controls were adequate and executed.
- Ensure that actual practice is in line with the SOD declaration.